

LABORATORY REPORT 1

JavaScript Basics & DOM Manipulation

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Section: BSIS 2A

Assignment Type: Laboratory Assignment

Connected Lab Activity: Week 4 Laboratory Guide

1. Objective

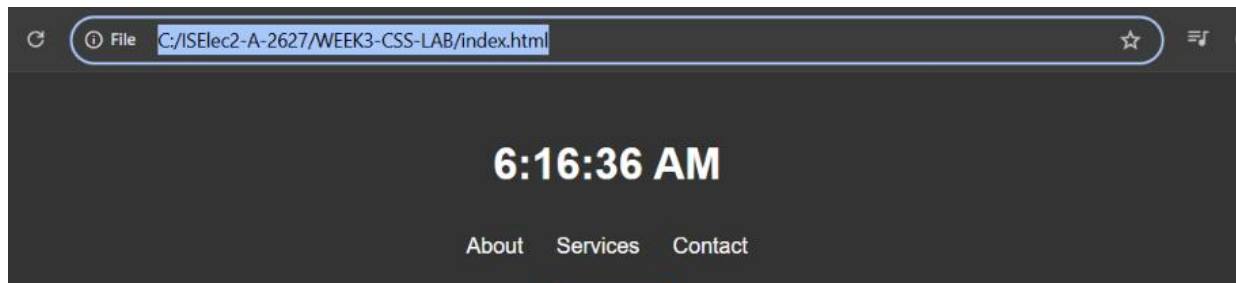
The objective of this laboratory activity was to extend the Week 3 web project by adding JavaScript interactivity and DOM manipulation while keeping the new features blended into the existing semantic HTML structure. The activity focused on a dynamic header clock, dark/light theme switching with localStorage, a dynamic to-do list, a keyboard event, and form validation using a modal popup.

2. Laboratory Requirements

Feature	Expected Behavior
Dynamic Header Update	Display the current time in the h1 and update it every second.
Theme Toggle	Switch between dark and light modes and save the selected theme in localStorage.
Dynamic To-Do List	Allow users to add tasks and remove each task dynamically.
Keyboard Event	Change the webpage background when the specified keyboard key is pressed.
Form Validation with Modal	Check the email for "@" and display either an error or success message in a modal.

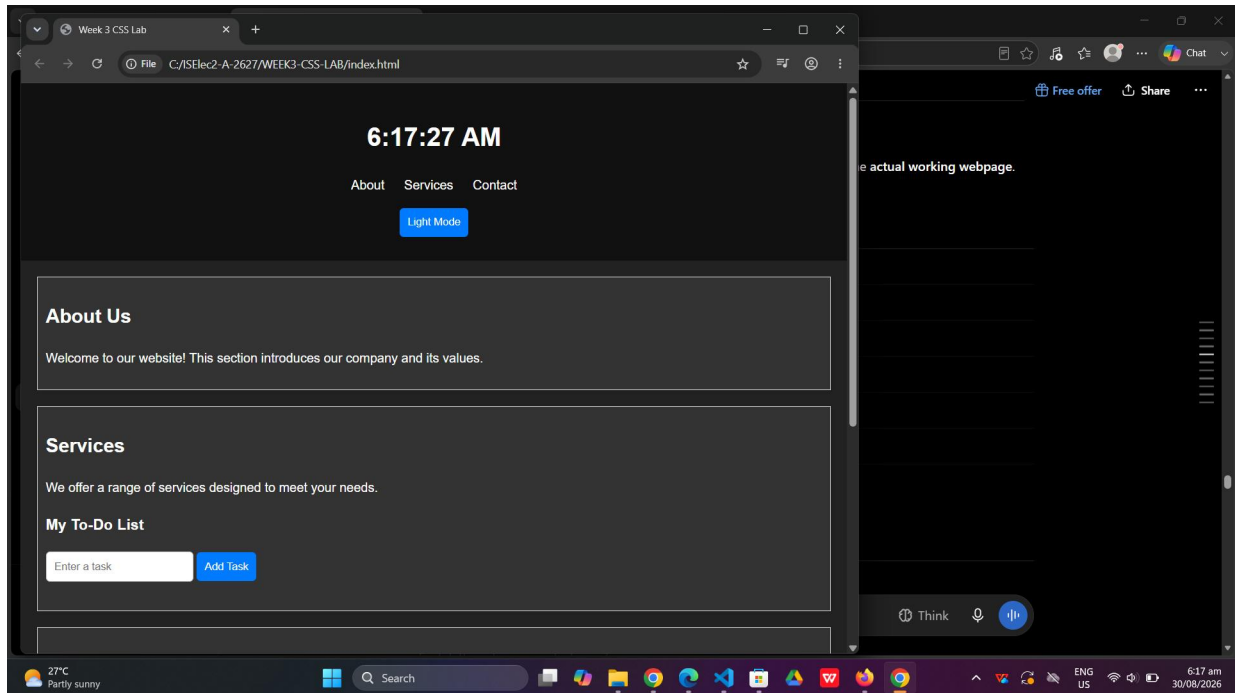
3. Screenshots and Descriptions

Screenshot 1 – Dynamic Header Update



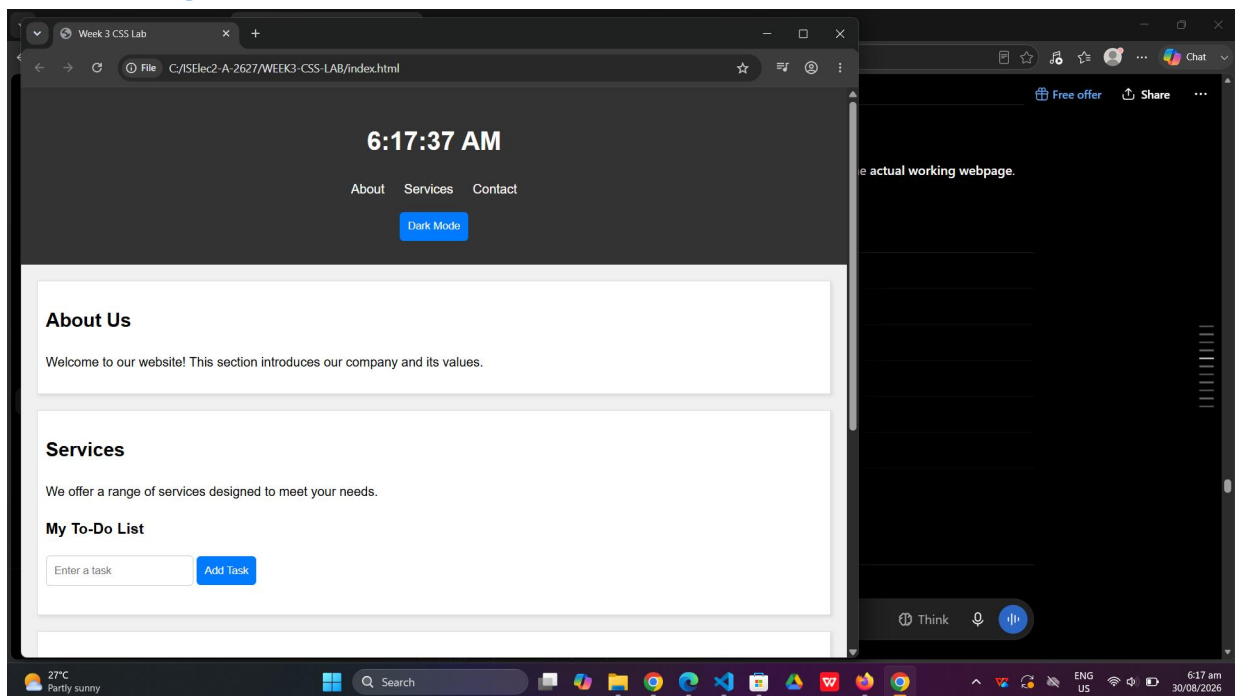
Description: This screenshot shows the webpage header displaying the current time, "6:16:36 AM." The time is shown in the h1 area of the semantic header. This demonstrates the Dynamic Header Update requirement, where the displayed time is updated continuously using JavaScript.

Screenshot 2 – Dark Mode



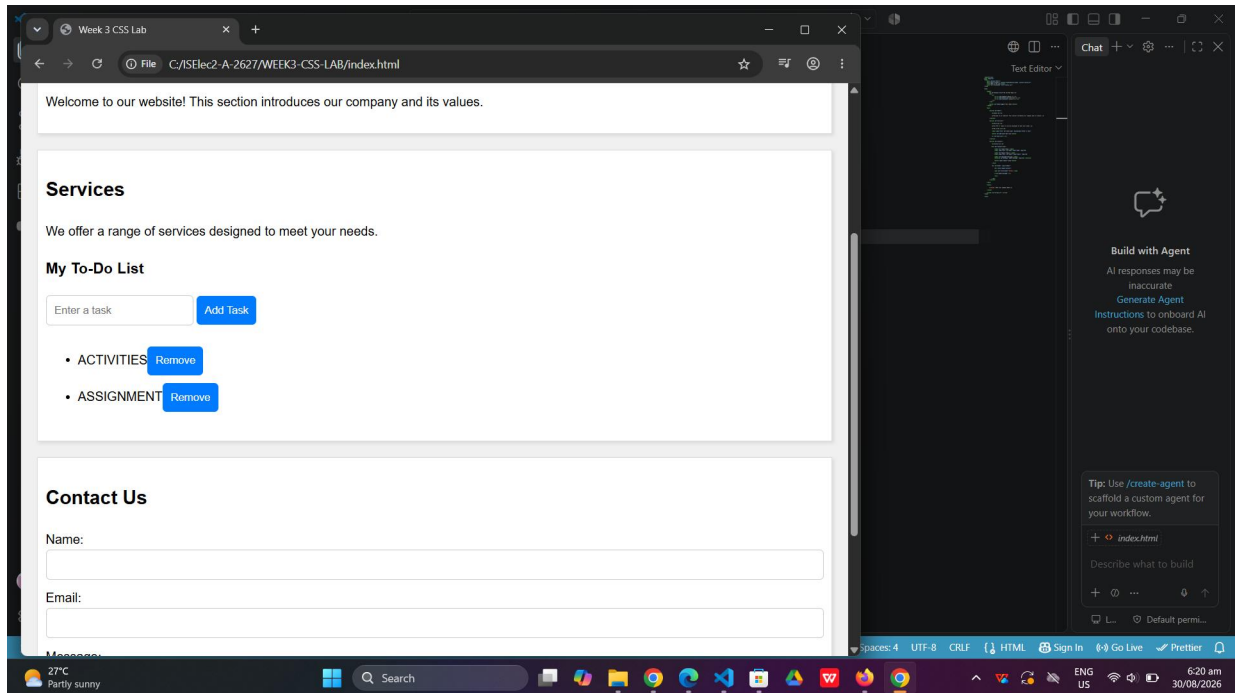
Description: This screenshot shows the webpage in dark mode. The header, sections, and text have changed to a darker appearance, and the button displays “Light Mode,” allowing the user to switch back. This demonstrates the Dark/Light Mode feature. The theme setting is also designed to be stored in localStorage so the selected theme can remain after refreshing the page.

Screenshot 3 – Light Mode



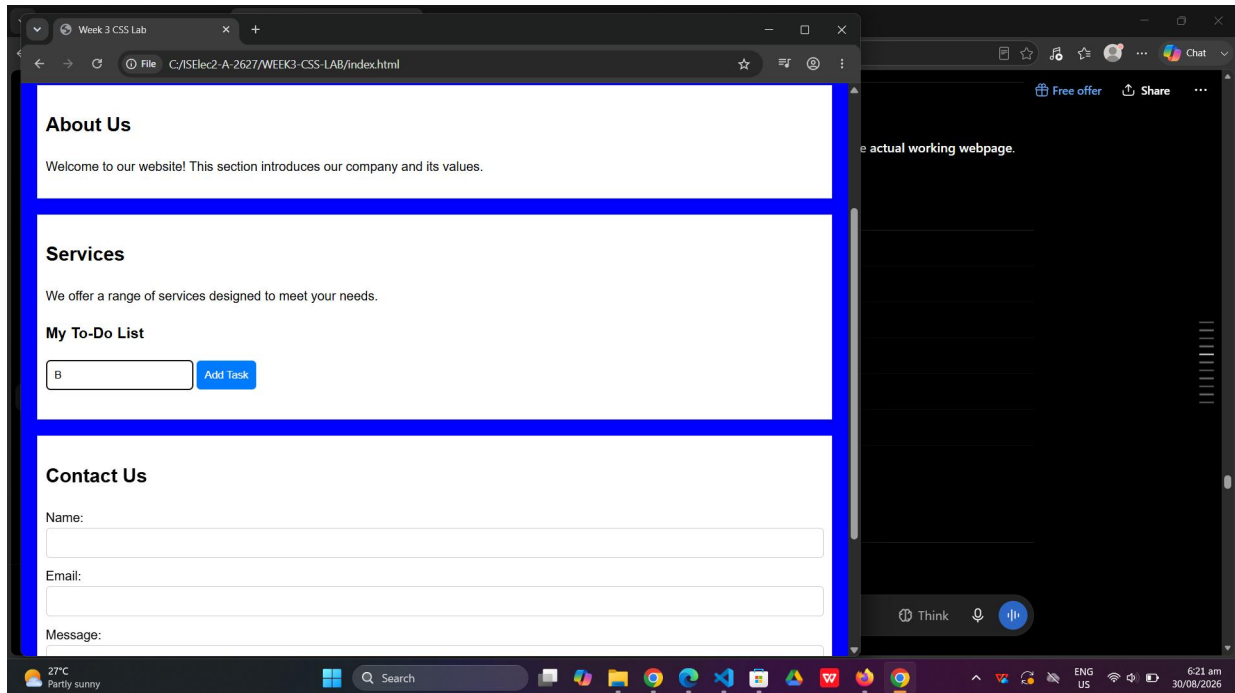
Description: This screenshot shows the webpage in light mode. The background and content sections are displayed with the light theme, and the button displays “Dark Mode.” This demonstrates that the user can switch between the two themes using the theme toggle button.

Screenshot 4 – Dynamic To-Do List with Remove Buttons



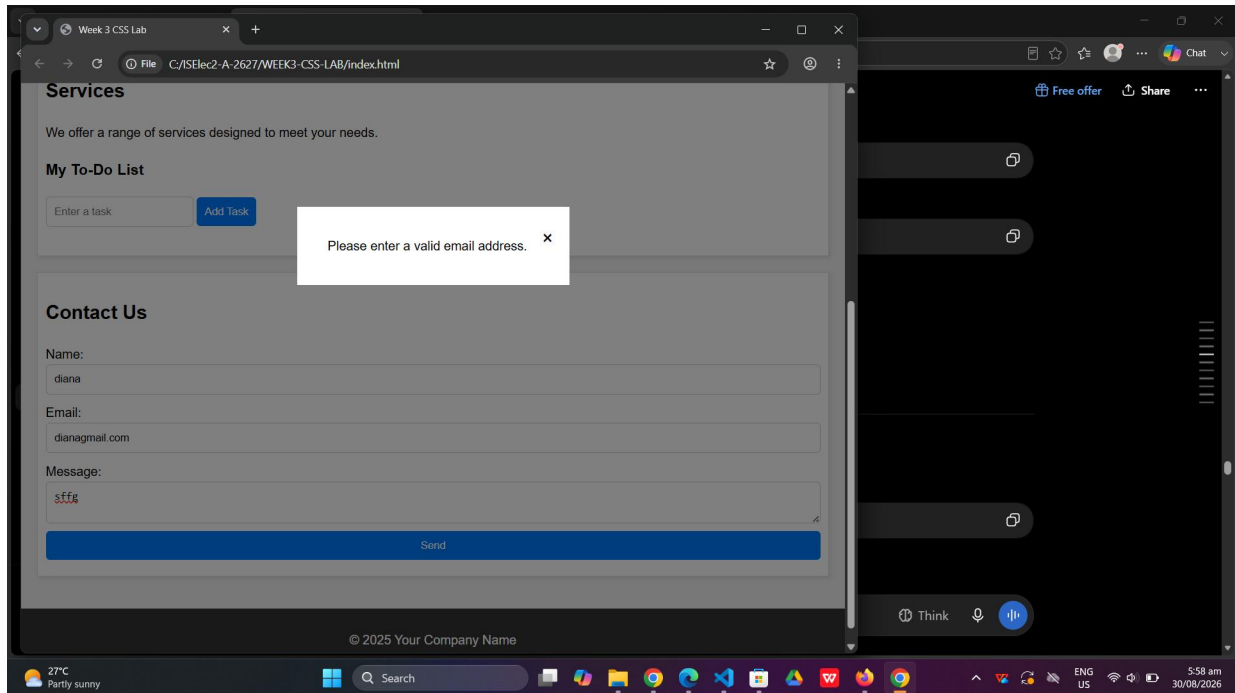
Description: This screenshot demonstrates the Dynamic To-Do List feature. The user has entered tasks such as “ACTIVITIES” and “ASSIGNMENT,” and each task appears in the list with its own Remove button. This shows that JavaScript is creating list items dynamically and allowing individual tasks to be deleted.

Screenshot 5 – Keyboard Event / Blue Background



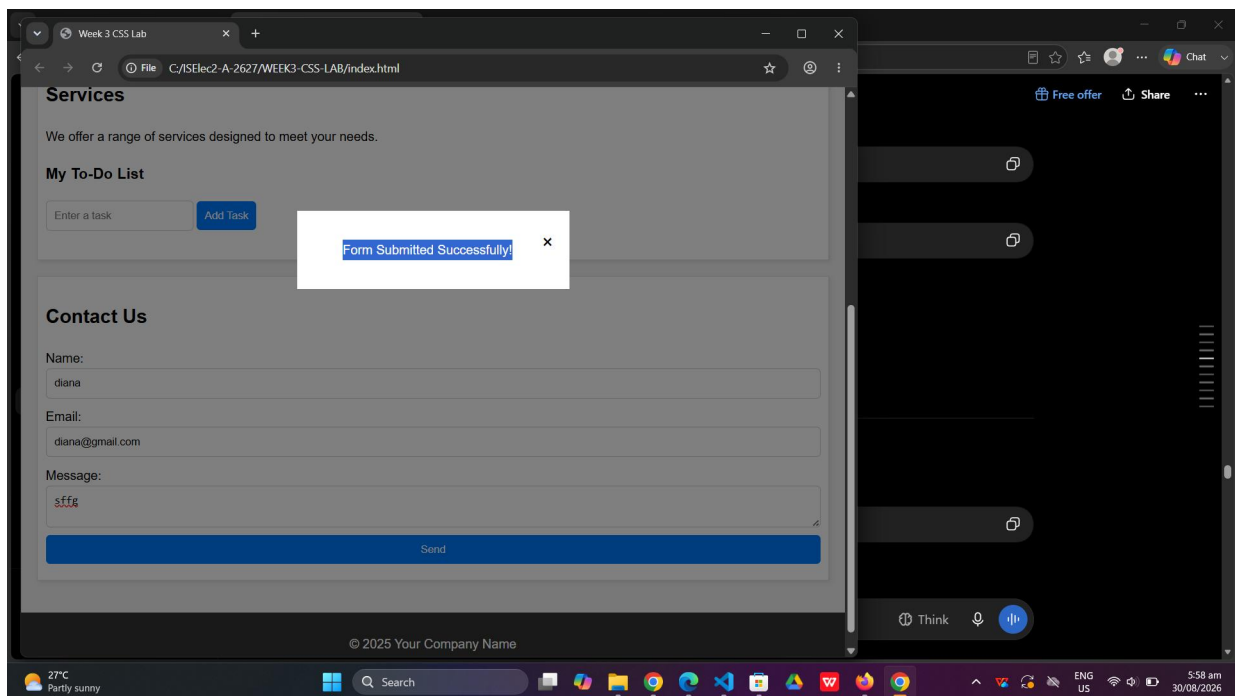
Description: This screenshot demonstrates the keyboard event requirement. The webpage background has changed to blue after the specified keyboard action. This confirms that JavaScript is listening for a keyboard event and changing the page appearance when the required key is pressed.

Screenshot 6 – Invalid Email Modal Validation



Description: This screenshot shows the form validation when an invalid email is entered. The email shown does not contain the required “@” symbol. Instead of using a normal browser alert, the webpage displays a modal popup containing “Please enter a valid email address.” This demonstrates that the form checks the email before accepting the submission.

Screenshot 7 – Successful Form Submission Modal



Description: This screenshot shows the successful form validation result. The email field contains “diana@gmail.com,” which includes “@.” After the form is submitted, a modal popup displays “Form Submitted Successfully!” This confirms that valid input produces the required success message inside the modal.

4. Results

The seven screenshots document the working JavaScript features added to the Week 3 webpage. The evidence shows the live header time, both dark and light themes, the dynamic task list with Remove buttons, the keyboard-triggered blue background, and both invalid and successful form-validation modal messages.

Feature	Result
Dynamic Header Update	Implemented and demonstrated
Theme Toggle	Implemented and demonstrated
Dynamic To-Do List	Implemented and demonstrated
Keyboard Event	Implemented and demonstrated
Form Validation with Modal	Implemented and demonstrated

5. Reflection (300 Words)

This laboratory activity helped me understand how JavaScript can make a webpage more interactive and responsive. One challenge I encountered was connecting the JavaScript code to the correct HTML elements. I had to make sure that the IDs used in the HTML matched the IDs selected in the JavaScript. When they did not match, buttons or other features would not work as expected.

Another challenge was implementing dark and light mode while also using localStorage. I learned that changing the appearance of a page is not enough because the selected theme must also be saved and loaded when the page is refreshed. This helped me understand how localStorage can store user preferences in the browser.

Creating the dynamic to-do list was useful because I learned how JavaScript can create new HTML elements while the webpage is running. I learned how to read the value from an input field, create a list item, add a Remove button, and delete the task when the button is clicked.

The keyboard event taught me how JavaScript can listen for actions from the user. In this activity, pressing a specific key changed the page background color. I also learned how event listeners connect user actions to JavaScript functions.

For the contact form, I learned how to prevent normal form submission, check whether the email contains the required “@” symbol, and display a modal message. Instead of using an alert, the modal provides a more customized user experience.

Overall, I learned JavaScript concepts such as DOM manipulation, event listeners, functions, conditions, localStorage, timers, input values, and dynamically creating elements. The activity was challenging at first, but testing each feature separately helped me understand how the different parts of JavaScript work together. It also showed me how JavaScript can improve the usability and interactivity of a webpage.

6. Conclusion

The laboratory activity successfully extended the Week 3 webpage with JavaScript-based interactivity. The activity demonstrated DOM manipulation, event handling, timers, browser storage, dynamic element creation, keyboard events, and form validation. The completed features made the webpage more interactive and user-friendly while preserving the semantic structure of the original project.